

## CORTEZ 8 SE, CO, USA

WMO: 745230

Lat: 37.2553N

Lon: 108.5035W

Elev: 8034

StdP: 10.90

Time Zone: -7.00 (NAM)

Period: 05-19

WBAN: 03061

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |             | Humidification DP/MCDB and HR |            |             |             |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|------------|-------------|-------------------------------|------------|-------------|-------------|------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |            |             | 99.6%                         |            |             | 99%         |            |             | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%      | 99%         | DP                            | HR         | MCDB        | DP          | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>7.6 | (c)<br>11.2 | (d)<br>-8.7                   | (e)<br>4.6 | (f)<br>24.9 | (g)<br>-4.5 | (h)<br>5.8 | (i)<br>26.9 | (j)<br>12.6           | (k)<br>25.3 | (l)<br>10.9 | (m)<br>26.1 | (n)<br>4.8            | (o)<br>N/A | (p)<br>0.467 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>21.0            | (c)<br>83.9     | (d)<br>53.7 | (e)<br>81.7 | (f)<br>53.2 | (g)<br>79.4 | (h)<br>52.6 | (i)<br>58.4         | (j)<br>73.3 | (k)<br>57.3 | (l)<br>73.1 | (m)<br>56.3 | (n)<br>72.0 | (o)<br>5.0           | (p)<br>N/A |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 54.3                            | 85.0 | 60.7 | 52.2 | 78.6 | 60.9 | 50.4 | 73.5 | 61.7 | 29.2          | 73.5 | 28.3 | 73.1 | 27.5 | 72.3 | 63.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |    |    |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|----|----|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |    |    |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      | DB | WB |
| (a) 11.0          | (b) 9.8 | (c) 8.7 | (e) 1.8                    | (f) 88.3 | (g) 4.3            | (h) 1.7 | (i) -1.2                                           | (j) 89.6 | (k) -3.7   | (l) 90.5 | (m) -6.1   | (n) 91.5 | (o) -9.2   | (p) 92.7 |    |    |
| (a) 11.0          | (b) 9.8 | (c) 8.7 | (e) 1.4                    | (f) 60.7 | (g) 4.7            | (h) 0.8 | (i) -2.0                                           | (j) 61.3 | (k) -4.7   | (l) 61.8 | (m) -7.3   | (n) 62.2 | (o) -10.7  | (p) 62.8 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 47.8   | 27.6  | 30.0  | 38.1  | 44.5  | 52.7  | 66.1  | 69.3  | 67.4  | 61.0  | 48.7  | 38.6  | 28.0 |
|                                                                                  | DBStd       | 16.82  | 7.53  | 8.57  | 8.76  | 8.72  | 8.58  | 6.42  | 4.13  | 4.37  | 6.67  | 8.49  | 9.15  | 8.55 |
|                                                                                  | HDD50       | 3078   | 694   | 559   | 377   | 204   | 76    | 2     | 0     | 0     | 10    | 125   | 348   | 683  |
|                                                                                  | HDD65       | 6665   | 1159  | 979   | 833   | 616   | 383   | 64    | 10    | 27    | 148   | 506   | 792   | 1148 |
|                                                                                  | CDD50       | 2259   | 0     | 0     | 9     | 38    | 160   | 484   | 599   | 540   | 339   | 84    | 6     | 0    |
|                                                                                  | CDD65       | 372    | 0     | 0     | 0     | 0     | 2     | 96    | 145   | 102   | 27    | 0     | 0     | 0    |
|                                                                                  | CDH74       | 2267   | 0     | 0     | 0     | 0     | 13    | 628   | 948   | 549   | 128   | 1     | 0     | 0    |
|                                                                                  | CDH80       | 371    | 0     | 0     | 0     | 0     | 0     | 101   | 194   | 68    | 8     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 4.5    | 4.7   | 5.0   | 5.0   | 5.4   | 4.7   | 4.4   | 3.9   | 3.7   | 4.0   | 4.5   | 4.5   | 4.5  |
| Precipitation                                                                    | PrecAvg     | 15.3   | 1.3   | 1.4   | 1.2   | 1.1   | 1.0   | 0.5   | 1.3   | 1.8   | 1.7   | 1.5   | 1.3   | 1.1  |
|                                                                                  | PrecMax     | 20.5   | 4.0   | 4.4   | 4.1   | 2.6   | 4.6   | 2.2   | 2.2   | 3.7   | 3.8   | 3.7   | 2.8   | 3.2  |
|                                                                                  | PrecMin     | 9.6    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.5   | 0.4   | 0.0   | 0.0   | 0.1  |
|                                                                                  | PrecStd     | 2.8    | 1.0   | 0.9   | 0.8   | 0.7   | 1.0   | 0.5   | 0.6   | 0.8   | 0.9   | 0.9   | 0.6   | 0.7  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 47.1  | 51.6  | 63.1  | 68.7  | 76.1  | 86.1  | 87.3  | 84.4  | 81.0  | 70.8  | 60.3  | 49.0 |
|                                                                                  |             | MCWB   | 34.5  | 37.1  | 41.7  | 43.6  | 46.8  | 52.4  | 54.9  | 53.5  | 52.8  | 47.5  | 41.4  | 34.5 |
|                                                                                  | 2%          | DB     | 42.5  | 48.2  | 58.6  | 64.2  | 72.3  | 83.0  | 84.4  | 82.0  | 77.6  | 66.3  | 57.0  | 44.8 |
|                                                                                  |             | MCWB   | 31.6  | 34.9  | 39.8  | 42.2  | 45.9  | 52.2  | 54.5  | 53.4  | 51.1  | 45.1  | 39.9  | 33.3 |
|                                                                                  | 5%          | DB     | 39.5  | 44.7  | 54.4  | 61.2  | 69.2  | 80.3  | 82.2  | 79.8  | 74.8  | 63.6  | 53.8  | 41.7 |
|                                                                                  |             | MCWB   | 30.3  | 33.5  | 37.4  | 41.1  | 44.9  | 50.8  | 54.3  | 53.1  | 50.1  | 44.0  | 37.9  | 31.5 |
|                                                                                  | 10%         | DB     | 37.0  | 41.7  | 50.8  | 58.0  | 66.2  | 77.8  | 79.6  | 77.3  | 71.7  | 60.3  | 50.4  | 39.2 |
|                                                                                  |             | MCWB   | 29.1  | 31.6  | 35.6  | 39.3  | 43.6  | 49.2  | 54.1  | 53.0  | 49.3  | 42.8  | 36.4  | 30.5 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 36.4  | 39.2  | 42.4  | 45.9  | 49.7  | 57.2  | 60.5  | 59.6  | 57.1  | 51.5  | 43.3  | 37.6 |
|                                                                                  |             | MCDB   | 43.6  | 50.9  | 61.1  | 65.4  | 70.8  | 78.8  | 73.8  | 71.2  | 70.6  | 62.9  | 56.9  | 44.7 |
|                                                                                  | 2%          | WB     | 34.3  | 36.4  | 40.7  | 43.6  | 47.3  | 55.0  | 58.8  | 58.3  | 55.0  | 49.0  | 40.8  | 35.6 |
|                                                                                  |             | MCDB   | 39.0  | 45.6  | 57.6  | 62.1  | 66.4  | 76.6  | 74.4  | 71.2  | 68.3  | 59.8  | 51.7  | 40.7 |
|                                                                                  | 5%          | WB     | 32.6  | 34.9  | 38.7  | 41.9  | 45.9  | 53.4  | 57.6  | 57.1  | 53.4  | 46.6  | 38.9  | 33.9 |
|                                                                                  |             | MCDB   | 37.7  | 43.5  | 53.4  | 58.5  | 65.6  | 74.7  | 74.3  | 71.1  | 68.1  | 58.2  | 49.6  | 39.4 |
|                                                                                  | 10%         | WB     | 30.7  | 33.2  | 36.7  | 40.3  | 44.4  | 51.8  | 56.8  | 55.9  | 52.2  | 44.9  | 37.1  | 32.2 |
|                                                                                  |             | MCDB   | 36.2  | 41.4  | 50.0  | 56.0  | 63.5  | 72.6  | 73.3  | 70.5  | 66.3  | 57.3  | 48.1  | 38.2 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 11.3  | 12.4  | 15.6  | 19.2  | 19.2  | 19.8  | 21.0  | 19.5  | 18.4  | 16.9  | 14.5  | 11.7 |
|                                                                                  |             | MCDBR  | 12.8  | 14.2  | 17.9  | 20.5  | 20.5  | 21.0  | 22.8  | 20.9  | 19.7  | 18.0  | 16.0  | 13.1 |
|                                                                                  |             | MCWBR  | 8.2   | 8.2   | 9.2   | 9.9   | 9.6   | 9.2   | 8.0   | 7.9   | 8.1   | 8.9   | 8.5   | 7.9  |
|                                                                                  | 5% WB       | MCDBR  | 11.7  | 13.2  | 17.2  | 19.7  | 20.1  | 19.7  | 21.3  | 19.5  | 18.5  | 16.1  | 15.7  | 11.7 |
|                                                                                  |             | MCWBR  | 7.9   | 7.7   | 9.2   | 9.6   | 9.6   | 9.4   | 7.7   | 8.0   | 8.2   | 8.4   | 8.6   | 7.8  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.222  | 0.227 | 0.244 | 0.265 | 0.277 | 0.282 | 0.315 | 0.315 | 0.280 | 0.259 | 0.231 | 0.223 |      |
|                                                                                  | taud        | 2.638  | 2.632 | 2.598 | 2.552 | 2.559 | 2.528 | 2.495 | 2.471 | 2.590 | 2.620 | 2.652 | 2.635 |      |
|                                                                                  | Ebn at Noon | 320    | 329   | 328   | 324   | 318   | 315   | 304   | 303   | 311   | 311   | 313   | 312   |      |
|                                                                                  | Edh at Noon | 21     | 25    | 28    | 31    | 32    | 33    | 34    | 34    | 28    | 25    | 21    | 20    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 906    | 1179  | 1644  | 2046  | 2348  | 2622  | 2307  | 2053  | 1796  | 1375  | 1018  | 784   |      |
|                                                                                  | RadStd      | 81     | 100   | 125   | 109   | 168   | 132   | 115   | 77    | 74    | 105   | 71    | 64    |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Regional (13 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
|                         | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page